

Setup and Calculation

We solve the compressible Euler equations in Lagrangian coordinates.

$$\partial_t \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \mathbf{u}^T + p \mathbf{I} \\ (E + p) \mathbf{u} \end{pmatrix} = 0$$

IGR Pressure

When solving shocks or mixing simulations, the use of artificial viscosity decreases the time step and washes out macro behavior. We add an igr pressure Σ to relieve these constraints.

$$p \mapsto p + \Sigma$$

$$\frac{\Sigma}{\rho} - \alpha \operatorname{div} \left(\frac{\nabla \Sigma}{\rho} \right) = \alpha \left(\operatorname{tr}([\mathbf{D}\mathbf{u}])^2 + \operatorname{tr}([\mathbf{D}\mathbf{u}]^2) \right)$$

Increasing the parameter α increases the amount of regularization.

Semidiscrete Laghos Equations

We build off an existing C++, MFEM-based library called Laghos. This solves the Euler equations in Lagrangian coordinates by treating the kinematic variables ($\mathbf{x}$, $\mathbf{v}$) as continuous finite elements and the thermodynamic variables (e , ρ) as discontinuous elements. It uses a slightly different form of the equations. Here, we modify the stress tensor from the usual equation of state.

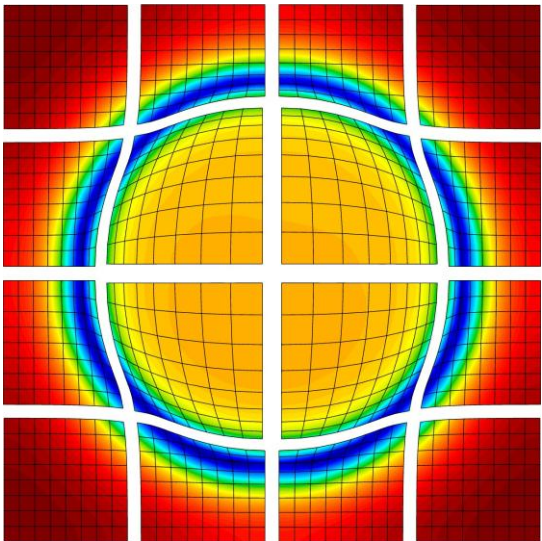
$$\rho \frac{dv}{dt} = \nabla \cdot \sigma, \quad \frac{1}{\rho} \frac{d\rho}{dt} = -\nabla \cdot \mathbf{v}, \quad \rho \frac{de}{dt} = \sigma : \nabla \mathbf{v}, \quad \frac{dx}{dt} = \mathbf{v}$$

Laghos Features

$$\sigma \mapsto \sigma + \Sigma \mathbf{I}$$

Laghos already has many desirable features built in and we are working to combine them with IGR.

- Parallelization with MPI and CUDA
- Visualization with GLVis and ParaView
- High order element spaces
- Partial assembly
- Curvilinear meshes
- Adaptive viscosity

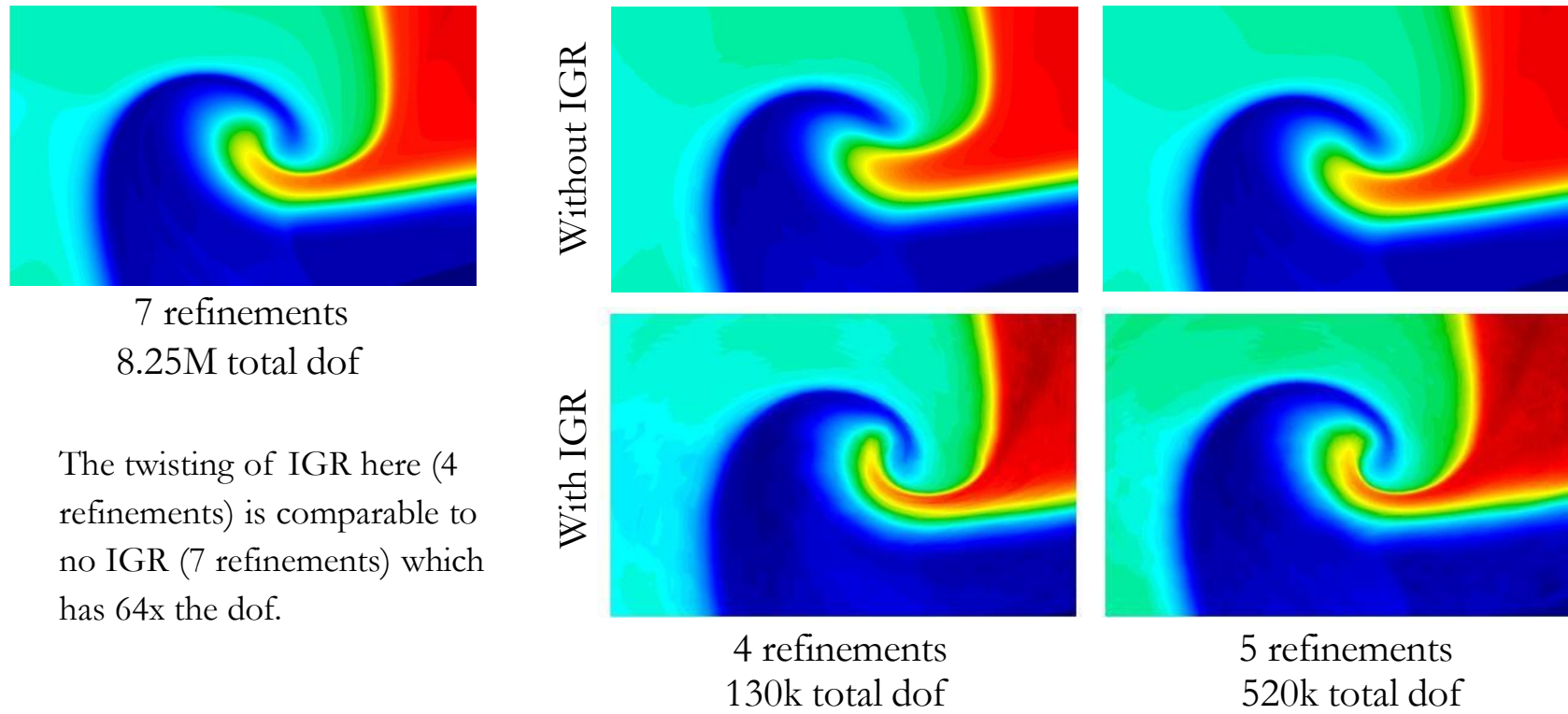


Results of Using IGR

Below we summarize some of the results of using IGR to regularize shocks and mixing problems. We find that using a constant α value under regularizes areas away from the shock and overregularizes regions near shocks due to the accumulation of mesh points about shocks. The most interesting results are for 2d problems so we have excluded the Shu-Osher results. A good choice of α is 4 times the estimate side length of an element.

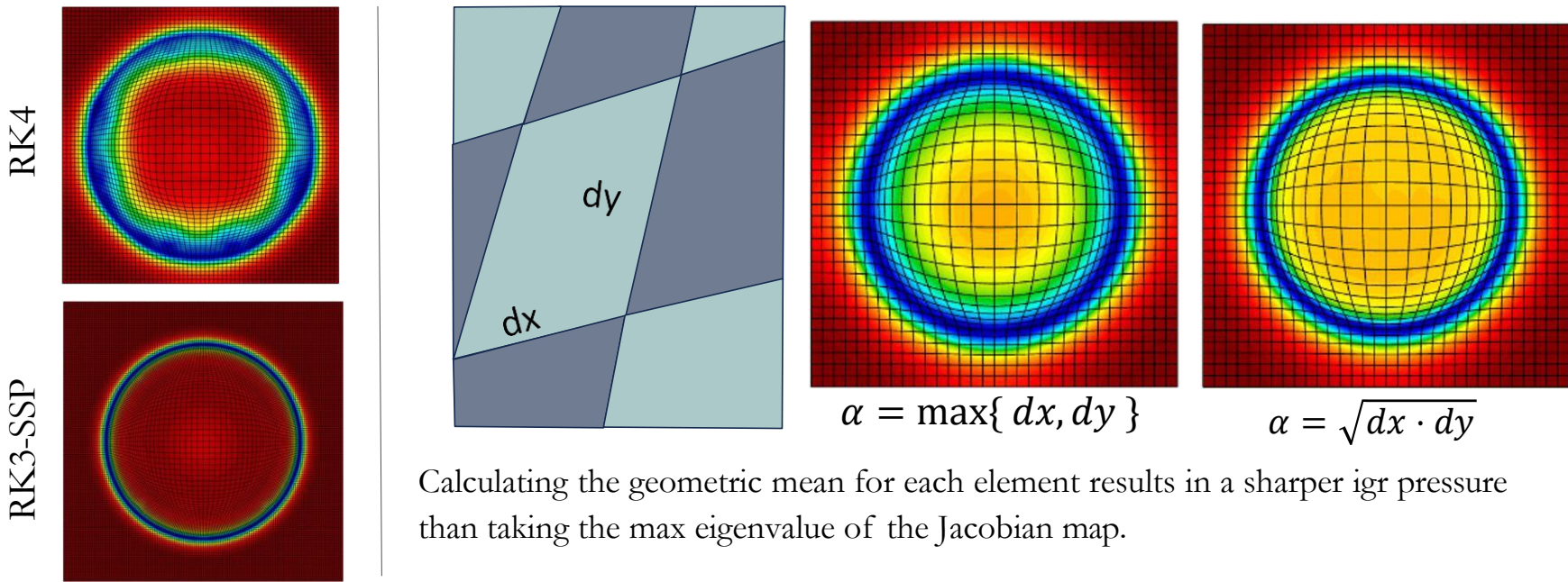
Triple Point

As we increase the resolution of the mesh, the runs converge to a physical solution. This can be seen in the triple point problem by twisting a greater amount with each refinement. We observe that at the same grid resolution IGR, does at better job at capturing the twisting compared to viscous regularization.



Sedov Blast

The Sedov Blast wave is where we test how far we can push our method. We examine the effect of choosing different time integrators and how we can choose to calculate α based on the size of each element.



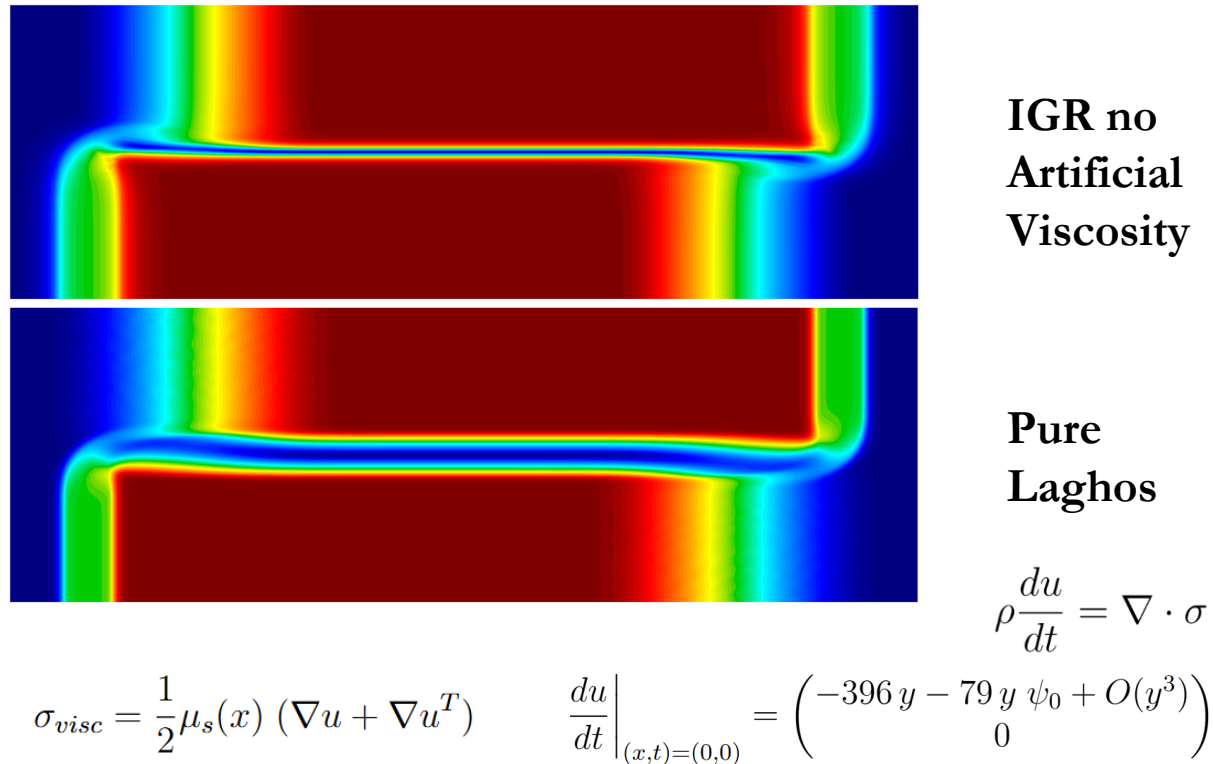
Additional Results

Runtime Behavior

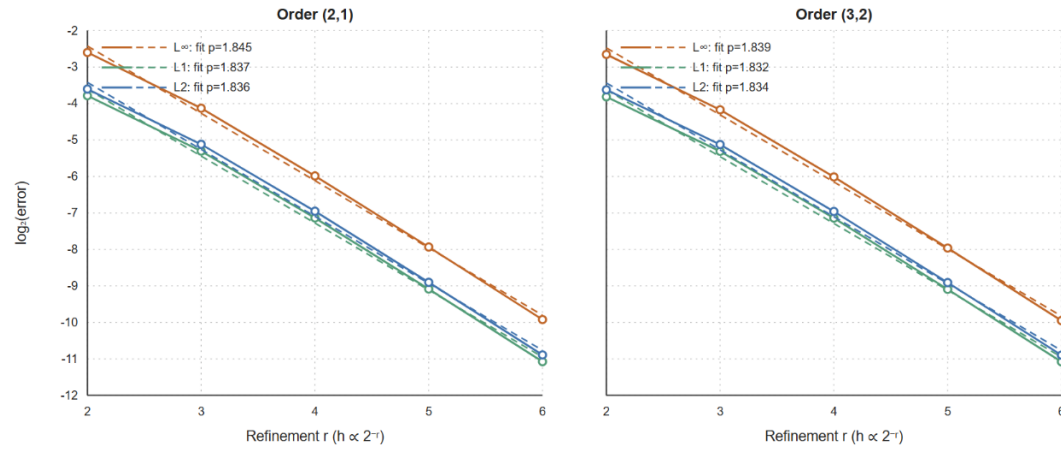
Being able to use fewer refinements for the same accuracy speeds up (for instance 66x faster in the example on the left). However, we also see comparing the 4 refinement runs that IGR uses 10x fewer time steps. We are working to speed up the added elliptic solve speed, but we find that 5 Jacobi preconditioned CG iterations seems to be sufficient.

$$\Delta t_q = \frac{CFL}{c_s/h_{min} + 2.5\mu_{visc}/\rho h_{min}^2}$$

Artificial Viscosity Widens Shearing Layers



Taylor Green has Second Order Convergence



Additional Runs

